

Cyclistic Bike Share Case Study

Business Analysis Report

1. Executive Summary

1.1 Business Problem

Cyclistic, a bike-sharing company based in Chicago, wants to increase the number of annual members by converting casual riders into subscribers. Understanding how casual riders and annual members use Cyclistic bikes differently is essential for developing an effective marketing strategy.

1.2 Objective

The objective of this analysis is to identify behavioral differences between annual members and casual riders using one year of trip data and provide data-driven recommendations to increase annual memberships.

1.3 Dataset

The analysis uses twelve months of Cyclistic trip data containing approximately 5.93 million rides.

1.4 Tools used

- R Programming
- Tableau
- Google Docs
- Google Slides
- GitHub

1.5 Business Task

Analyze the riding behavior of annual members and casual riders to answer the following business question.

How do annual members and casual riders use Cyclistic bikes differently, and how can these differences inform a strategy to convert casual riders into annual members?

1.6 Analysis Process

The project followed the six phases of the Google Data Analytics process:

- Ask
- Prepare
- Process
- Analyze
- Share
- Act

KPI	Result
Total Rides	5.93 Million
Member Rides	3.82 Million
Casual Rides	2.12 Million
Member Share	64.3%
Peak Hour	17:00 (5:00PM)
Preferred Bike	Electric Bike
Weekday Rides	4.25 Million

2. Key Findings

2.1 Member vs Casual Rider Behavior

Observation

Analysis of 5.93 million rides revealed clear behavioral differences between annual members and casual riders.

- Annual members completed 3.82 million rides (64.3%), while casual riders completed 2.12 million rides (35.7%).
- Casual riders had a significantly longer average ride duration than annual members.
- Annual members rode primarily during weekdays, indicating regular commuting behavior.
- Casual riders showed higher riding activity during weekends, suggesting recreational and leisure usage.
- Annual members generated a larger share of total rides despite spending less time per ride.

Business Insight

These findings indicate that annual members primarily use Cyclistic bikes for daily transportation, while casual riders mainly use the service for leisure and recreational purposes. Because casual riders already use the service frequently during weekends, they represent the strongest opportunity for conversion into annual members through targeted marketing campaigns.

2.2 Seasonal Trends

Observation

Monthly ride analysis showed a clear seasonal pattern.

- Ride volume increased steadily from spring through summer.
- The highest number of rides occurred during the summer months (June–September).
- Ride activity declined significantly during the winter months (November–February).
- Both rider groups followed similar seasonal trends, although members consistently recorded more rides than casual riders.

Business Insights

Cyclistic experiences strong seasonality. Marketing campaigns promoting annual memberships should be concentrated before and during the summer season when customer demand is highest and casual rider activity increases.

2.3 Rideable Type Preference

Observation

Rideable type analysis revealed that electric bikes were the most preferred bike type for both rider groups.

- Electric bikes recorded the highest number of rides (4.01 million).
- Classic bikes remained popular but were used less frequently (1.92 million).
- Members generated higher ride counts than casual riders across both bike types.

Business Insight

The popularity of electric bikes suggests that customers value convenience and comfort. Future promotional campaigns should emphasize the benefits of unlimited access to electric bikes through annual memberships.

2.4 Peak Riding Hours

Observation

Peak riding activity occurred at 17:00 (5:00 PM) for both annual members and casual riders.

- Member rides increased sharply during the morning commute and reached their highest level at 17:00.
- Casual riders also peaked at 17:00, although their ride pattern was more evenly distributed throughout the afternoon.
- Members consistently recorded higher ride counts than casual riders throughout the day.

Business Insight

The strong evening peak suggests that many annual members use Cyclistic bikes for commuting after work. Casual riders also show increased demand during the evening, making this an ideal time to promote membership offers through the mobile app, station advertisements, and digital notifications.

2.5 Weekday vs Weekend Analysis

Observation

Distinct riding patterns were observed between annual members and casual riders.

- Annual members generated most of their rides during weekdays.
- Casual riders showed relatively higher usage during weekends.
- Weekday rides totaled approximately 4.25 million, representing the majority of overall rides.

Business Insight

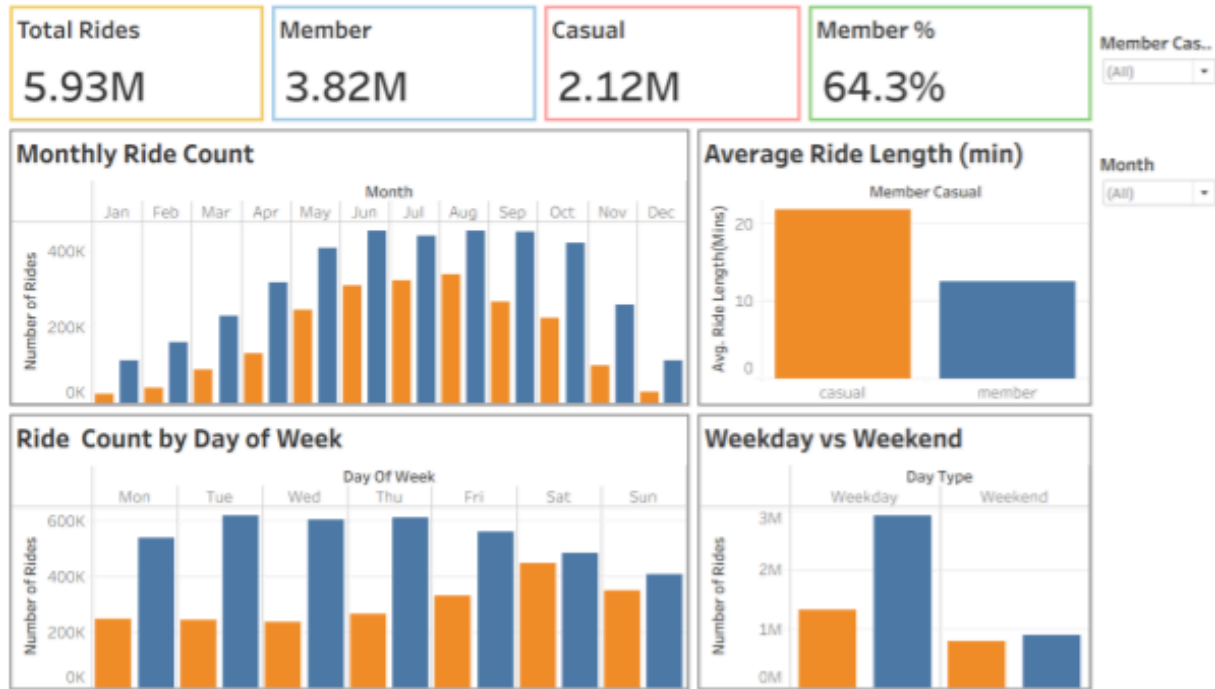
The difference in riding behavior indicates that members primarily rely on Cyclistic for daily transportation, while casual riders use the service mainly for leisure and recreation. Weekend-focused marketing campaigns and membership promotions could encourage casual riders to become annual members.

Note: Weekday volume is aggregated across 5 days, which is why it exceeds any single weekend day even though Saturday is the single busiest individual of the week.

- Dashboard 1: Executive Summary

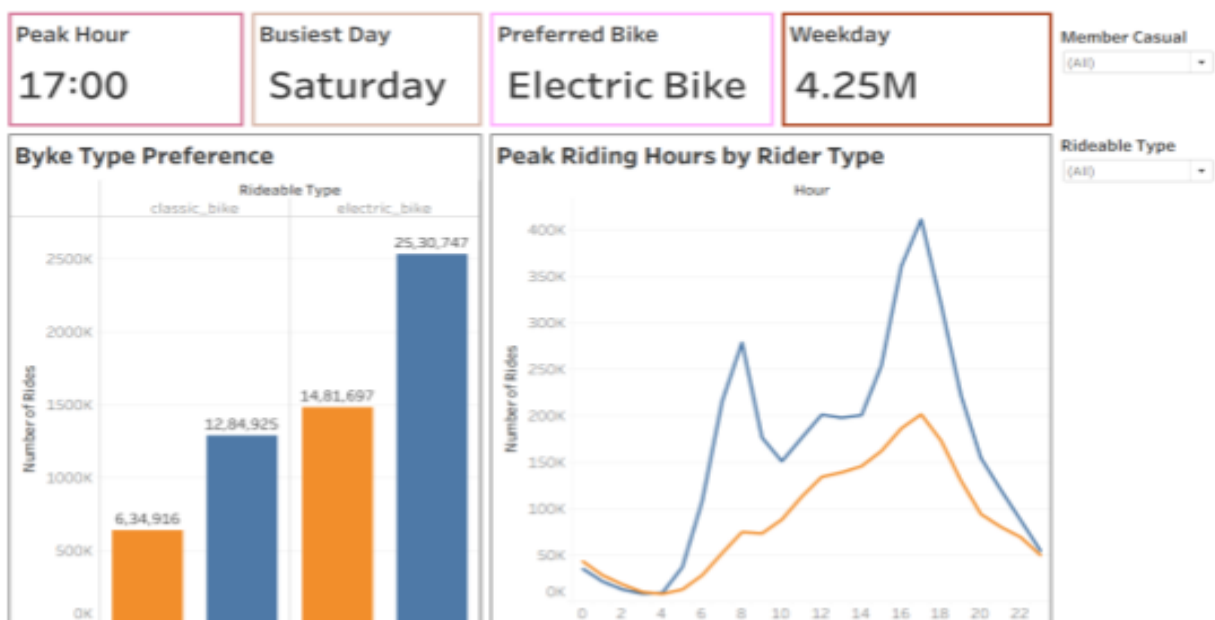
Cyclistic Bike Share Analysis Dashboard

Business Insights for Converting Casual Riders into Annual Members



- Dashboard 2: Customer Behavior Analysis

Cyclistic Customer Behavior Dashboard



3. Business Recommendation

Recommendation 1: Target Weekend Riders

Launch membership campaigns during weekends when casual rider activity is highest. Weekend users are the most promising audience for converting to annual memberships.

Recommendation 2: Promote Electric Bike Membership Benefits

Highlight unlimited access and cost savings associated with electric bikes, as they are the most preferred rideable type among both rider groups.

Recommendation 3: Launch seasonal Campaigns

Increase advertising and promotional activities before and during the summer months (June–September) when overall demand reaches its highest levels.

Recommendation 4: Focus on Evening Promotions

Display membership offers during the evening peak hour (17:00) through the Cyclistic mobile application and at high-traffic stations.

Recommendation 5: Create Commuter Membership Plans

Develop membership packages specifically designed for casual riders who may already be regular commuters, emphasizing cost savings, convenience, and unlimited riding benefits.

A Note on Scope

This analysis and its recommendations are based solely on ride count, ride timing, ride duration, and rideable type, as the dataset does not include pricing, revenue, or profitability information. Consequently, no revenue or dollar-based projections have been made to avoid unsupported conclusions.

Additionally, station-level geographic analysis has been intentionally excluded because the primary business objective is to understand behavioral differences between casual riders and annual members, rather than location-specific usage

patterns. A geographic analysis would require a separate research scope and business objective.

4. Conclusion

This analysis identified clear behavioral differences between annual members and casual riders. Annual members primarily use Cyclistic bikes for weekday commuting and complete more rides with shorter durations, while casual riders tend to take longer recreational rides, especially during weekends.

Based on these findings, Cyclistic can improve annual membership growth by focusing marketing efforts on casual weekend riders, promoting electric bike benefits, increasing summer campaigns, and targeting customers during peak riding hours.

These data-driven recommendations support the company's goal of converting more casual riders into annual members. If implemented, they could help improve member acquisition and customer retention through data-driven marketing strategies.

Future Work

Future analysis could incorporate station-level geographic data, customer segmentation, pricing information, and revenue metrics to evaluate profitability and develop a more targeted membership conversion strategy.

Project Limitations

This analysis was completed using R for data cleaning and Tableau for visualization. SQL analysis was not included because the cleaned dataset (approximately 1.42 GB and 5.93 million records) exceeded the practical import capabilities of the available local SQL workflow. The project instead focuses on business insights derived from the complete cleaned dataset.

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